Synthetic Grid-Dynamics Pretraining for Motor-Speech Decoding from Acute μECoG

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Outline

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The Data Bottleneck

Acute intraoperative μECoG:

- ~1 min utterance / patient (~150 trials)
- 128–256 channels at 200 Hz, HGA z-score
- 8–12 patients, left sensorimotor cortex

What we know:

- Per-patient CE: **PER = 0.700** (S14 best)
- Cross-patient LOPO: PER = 0.846 (near chance)
- H=128 backbone: 55× train/val gap
- CTC suffers blank collapse (95% blank ratio)

[Spalding 2025; Duraivel 2023, 2025; Singh 2025]

Key numbers

Patients	8–12
Trials/pt	46–178
Sig channels	63–201
Phonemes	9 (3/trial)
Utterance	~1 min/pt
Total pooled	~3 hrs

Bottleneck

Input compression (pool → 64 dim) destroys spatial resolution, not GRU capacity.

Why Not Existing SSL?

Spike-based foundation models

BIT (367 hrs), NDT3 (2000 hrs)

- Poisson spikes ≠ Gaussian HGA
- Orders of magnitude more data

ECoG SSL

wav2vec ECoG (CPC, not 2.0)

- Min successful: ~30 min/pt
- We have ~1 min/pt (30× smaller)
- 1/12 cross-pt transfers worked

What transfers without weights:

- Architecture patterns (TimeSformer, wav2vec)
- Masking strategies (VideoMAE)
- Per-patient layer design (Singh, Levin)

Opportunity

Rigid 2D grid structure is *known a priori*—we can synthesize unlimited spatiotemporal movies with controlled inductive biases.

[Zhang 2026 (BIT); wav2vec ECoG; VideoMAE (Tong 2022); Singh 2025; Levin 2026]

Core Idea: Three-Stage Pipeline

Hypothesis: Synthetic spatiotemporal dynamics teach an encoder to process 2D grids evolving over time. This transfers to real cortical activity.

Analogy: Both synthetic generators and μECoG produce 2D grids evolving over time; the model must extract spatial patterns and track temporal evolution.

Three Claims

1. **Does SSL pretraining help** for acute intra-op μECoG motor-speech decoding at <200 trials/patient?
   - Test: Method A/B vs D (supervised baseline)
2. **Does synthetic pre-pretraining add value** beyond real-data-only SSL?
   - Test: Method A vs B (neural-only); Method J controls for step count
3. **Does structured synthetic dynamics help** beyond matched-statistics non-dynamic exposure?
   - Test: Method A vs K (destroyed-dynamics control)

Primary endpoint

Full-window readout $[-0.5, +1.0\text{s}]$ with per-position CE—validated at PER = 0.700.
Motor-focused $[-0.15, +0.6\text{s}]$ is preregistered mechanistic secondary.

Generator Ladder: Comparative Hierarchy

Lv	Generator	Local coup.	Nonlinear	Diverse	New property
0	Smooth Gaussian AR				*Baseline*: smooth, correlated
1	Switching LDS	✓			Regime switching
2	Damped Wave	✓			Oscillatory local coupling
3	Gray–Scott	✓	✓		Nonlinearity (RD)
4	FitzHugh–Nagumo	✓	✓		Excitable-medium dynamics
5	NCA Random MLP	✓	✓	✓	Rule diversity

Key comparisons:

- L0 vs L1: regime switching?
- L1 vs L2: oscillatory coupling?
- L2 vs L3–4: nonlinearity?
- L4 vs L5: rule diversity?
- L0 vs Mixed: everything together?

If Mixed wins, the paper becomes "diverse synthetic dynamics help"—not "cortex-like dynamics help." Both interpretations prepared.

Also: Sequential hotspot activation (Gaussian bump on random trajectory) tests trajectory vs. local-coupling structure.

Generator Ladder: Confounds & Calibration

Confound acknowledgment

Each level changes a primary property *and* secondary characteristics (spectral shape, stationarity, event morphology). If L3 > L2, nonlinearity is *associated* with improvement, not strictly causal—Gray–Scott also changes spectral shape relative to the damped wave. Pairwise matched controls deferred. **Method K** provides an orthogonal cut: dynamics vs. statistics.

Coarse calibration (source-only)

1–2 params/generator to avoid degenerate dynamics:

- PSD: main power below 20 Hz
- Spatial correlation: 2–10 electrodes
- Kurtosis > 2
- Event sparsity: 5–30%

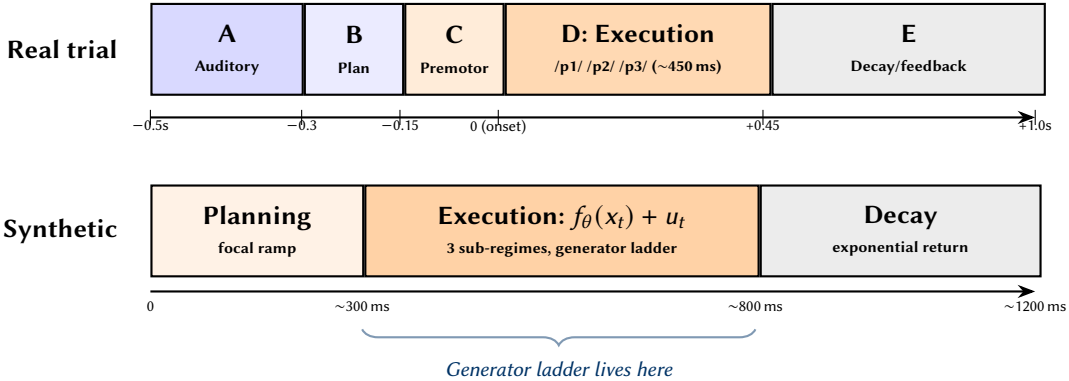
Not optimization—prevents blowup/collapse.

Matching statistics (source pts only)

Computed for each generator vs real uECOG:

- Temporal PSD (Welch, evoked + residual)
- Spatial covariance length scale
- Amplitude distribution (kurtosis)
- Dead-mask prevalence
- Event sparsity per trial phase
- RT distribution (log-normal fit)

Synthetic Trial Anatomy



Forced generators: $x_{t+1} = f_\theta(x_t) + u_t$ where u_t is phase-dependent drive. Sub-event templates are *generic* random focal patterns, NOT phoneme-specific.

Nuisance realism: z-normalize → IID noise ($\sigma \sim \mathcal{U}(0.3, 0.8)$) → correlated noise (rank 1–3, SD $\sim \mathcal{U}(0.2, 0.6)$) → dead electrode masks → anti-alias LPF → subsample to 20 Hz.

Masking Objective

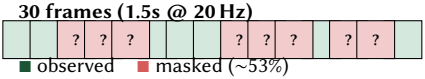
Masked temporal span prediction:

1. Mask 40–60% of frames (3–6 spans)
2. Predict masked frames from bidirectional context
3. Loss = MSE(predicted, actual) on masked frames only

Why aggressive masking?

At 20 Hz (30 frames/trial), the signal is smooth and temporally autocorrelated. 10–17% masking → trivially solvable by linear interpolation.

[VideoMAE (Tong 2022): aggressive masking critical for SSL on smooth signals]



Required baselines

1. Linear interpolation
2. Phase-template average
3. Patient-specific template
4. Cropped-window eval
5. Destroyed-dynamics (Method K)

Stage 2: Response-Locked Trial Adaptation

Honest supervision accounting

Stage 2 uses **no phoneme labels** but is **not unsupervised**. It requires response-onset timestamps (MFA), trial identity (every sample is a speech trial), and patient identity. This is **weakly-supervised trial-epoch adaptation**.

The shortcut problem: Response-locked trials have a stereotyped phase calendar (auditory → plan → execute → decay). The acoustic regression pivot showed 500ms-shifted controls were competitive—the model learned trial structure, not phoneme-sensitive mappings.

Anti-leakage defenses:

1. Full-epoch off-center crops

Load $[-1.0, 1.5\text{s}]$ (50 frames @ 20 Hz).
Crop to 30–40 frames with random offset.
Removes entire phases from some samples.

2. Time-reversal augmentation

10% probability: reverse temporal order.
Destroys causal phase ordering.
If it hurts → model relies on phase structure.

Transfer-proxy gate: Probe every 5K steps on locked dev patient. Earliest-within-1SE checkpoint rule. Source-only sensitivity analysis alongside.

Stage 2: Patient Pools & Diagnostics

Mode	Synth calibration	Stage 2 neural pool	Evaluation
Inductive (primary)	PS $\setminus\{d, t\}$	Unlabeled trials: PS $\setminus\{d, t\}$	Patient t
Transductive (secondary)	Same as inductive	Unlabeled trials: PS $\setminus\{d\}$ (incl. t)	Patient t

d = fixed dev patient (e.g., S26), t = target patient in current fold.

Required diagnostics

- Per-phase prediction loss
- Patient-ID vs phoneme-ID probes
- Content-collapse check:
  - Output entropy < 1.5 bits $\rightarrow$ alarm
  - Unigram KL from uniform
  - Stereotypy index $< 10\% \rightarrow$ alarm

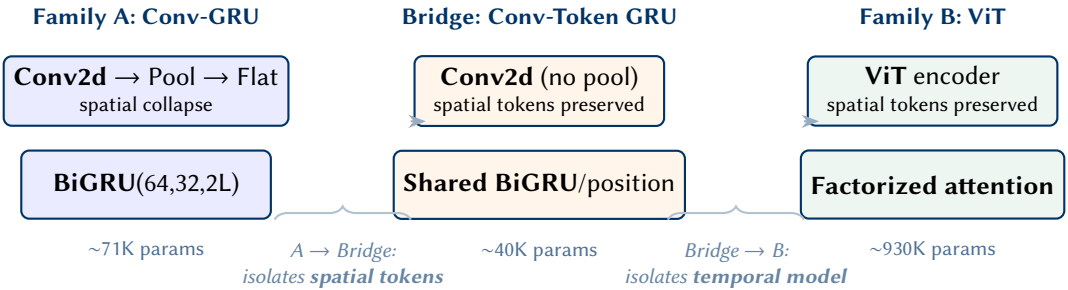
JEPA escalation triggers

Promote latent prediction if **any** of:

1. Event-frame gains vanish at execution-only decoding
2. Patient-ID $> 2\sigma$ without phoneme-ID gain
3. Pretext MSE plateaus >5K steps w/o transfer-proxy lift

[JEPA: Assran et al. 2023 (I-JEPA)]

Three Architecture Families



Conv-GRU (conservative): Preserves spatial-collapse bottleneck. If pretraining helps *despite* this → robust signal.

Conv-Token GRU (bridge): Per-position BiGRU, no cross-position interaction. Isolates spatial preservation.

ViT (high-capacity): Factorized space-time attention. Freezing required (~5.3K trainable at Stage 3).

[TimeSformer (Bertasius 2021); Singh 2025 (per-patient layers)]

Spatial Pooling & Positional Encoding

Spatial pooling (Family B & Bridge):

- **Mean-pool** (co-default): simple average
- **Top-k pool** (co-default, k=16): average k highest-norm tokens. Parameter-free
- **Learned attention** (Tier 1 ablation): query cross-attends to spatial tokens, +33K params

Rationale: Mean-pool erases focal articulatory activation on <2mm grids. Top-k is the parameter-free fix.

Positional encoding:

Local geometry PE (default)

PE = Linear($3 \to d$) on:

$$(\text{row} \times \text{pitch}_{\text{mm}},\ \text{col} \times \text{pitch}_{\text{mm}},\ \text{dead_frac})$$

Known pitches: 128ch = 1.33 mm, 256ch 12×22 = 1.72 mm.
8×32/8×34: surrogate 1.33 mm (flagged).

Not included: MNI anatomical encoding (future ablation—local mm coordinates do NOT encode where the array sits on cortex).

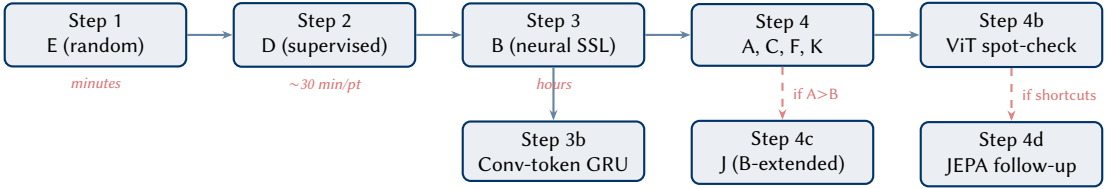
Augmentation consistency: Flips/rotations must transform PE inputs identically.

Comparison Methods (First Round)

	Method	Description	Tests
A	Structured Synth + Neural	Mixed pool {switching LDS, wave, GS, FHN, NCA, hotspot} $\rightarrow$ neural	Hypothesis
B	Neural Only	Masked span on real trials, random init	Claim 1
C	Smooth AR + Neural	Generic smooth synthetic $\rightarrow$ neural	Claim 2
D	Supervised Baseline	No pretraining, end-to-end CE	Baseline
E	Random-Init Frozen	Freeze random encoder, train head only	Noise floor
F	Random-Scaffold + Neural	Same generators, random phase count	Scaffold
K	Destroyed-Dynamics + Neural	Phase-randomized / time-shuffled	**Claim 3**
J	B-Extended (2$\times$ steps)	*Automatic if $A > B$ by bootstrap CI*	

Step-matched: A, B, C, F, K all get S_{total} optimizer steps.
Residual confound: A sees unlimited synthetic diversity; B reuses $\sim$1500 trials. Method J controls this.

Execution Order: Pilot-First Sequencing



Interpretation ladder (Step 4):

Outcome	Interpretation
A > K > C > B	Structured dynamics help beyond statistics (**strongest**)
A > F > C > B	3-event scaffold matters
A ≈ K > C > B	Spatial statistics help, not dynamics (narrower claim)
A ≈ C > B	Any smooth pretraining helps (weakest publishable)
A ≈ B ≈ D	SSL doesn't help (null result)

Stage 3: Per-Patient Fine-tuning

Default freeze configuration:
Freeze all except:

- CE classification head
- Per-patient Conv2d / patch embed
- Predictor LayerNorm parameters

Trainable: ~2.3K (conv-GRU), ~2.2K (bridge), ~5.3K (ViT)

Cross-validation:
Grouped-by-token (all reps of same CVC/VCV token in same fold). Fixed splits saved to JSON, reused across all methods.

Note: 0.700 PER used stratified CV—not directly comparable.

Readout windows:

1. **Full-window** $[-0.5, +1.0\text{s}]$
   headline primary, validated at PER = 0.700
2. **Motor-focused** $[-0.15, +0.6\text{s}]$
   preregistered mechanistic secondary
3. **Execution-only** $[0, +0.5\text{s}]$
   ablation, strictest test

Loss: 27-way CE (3 pos $\times$ 9 phonemes)
+ Three-bin temporal pooling (secondary)

Content-collapse checks

Entropy, unigram KL, stereotypy index.
LOPO CE showed 100% correct-length but only 21% position accuracy.

Statistical Framework

Patient-level aggregation:
Atomic unit = mean PER across CV folds per patient per method.

Population statistics:

- Per-patient table (all methods, all windows)
- Mean ± std of patient-level PER
- **Wilcoxon signed-rank** on paired PERs
- Bootstrap 95% CIs (10K resamples)
- Cohen's d + Holm–Bonferroni correction

Primary: A vs D, A vs B
Secondary: A vs C, A vs F, A vs K

Surrogate null control

For A vs D headline: permute phoneme labels within each patient (preserving patient/fold structure), retrain Stage 3 head, compute PER. 100× per patient.

If real improvement falls within null 95th percentile → gain is not label-informative.

Power considerations

N = 8–12 patients with noisy estimates. Bootstrap CIs + per-patient scatter plots + effect sizes supplement Wilcoxon.

Known Risks & Defenses

Risk	Mitigation	Evidence / Precedent
Synth → neural gap	Nuisance aug, coarse calibration, Method C/K	VideoMAE, I-JEPA: gap bridgeable
Trivial interpolation	40–60% masking, interp. baseline, event-frame diagnostic	VideoMAE: smooth signals need aggressive masking
Trial-calendar leakage	Off-center crops, time-reversal aug, per-phase loss	Acoustic regression: 500ms-shift competitive
Content collapse	Entropy/KL/stereotypy diagnostics	LOPO: 95% blank (CTC), 21% acc (CE)
ViT overfitting	Freeze all but ~5.3K; conv-GRU fallback	H=128: 55× train/val gap
Low power (N=8–12)	Bootstrap CIs, surrogate null, per-patient plots	Standard for intra-op BCI

Precautionary lessons: Data scarcity is the wall · Per-patient adaptation is the default assumption [Singh 2025] · Auxiliary targets need negative controls · MPS breaks with mixed grid sizes.

Decisions to Lock

1. **Headline evaluation contract**:
   Inductive Stage 2 primary · Full-window CE primary endpoint · Motor-focused secondary
2. **First-round architecture set**:
   Conv-GRU pilot → conv-token GRU bridge → ViT spot-check with paired adapter ablation
3. **Stage 2 interpretation language**:
   "Weakly supervised response-locked trial adaptation" (not "unsupervised neural SSL")
4. **Surrogate-pitch fallback**:
   1.33 mm for 8×32/8×34 until hardware specs verified?
5. **Dev patient selection**:
   Single fixed dev patient (e.g., S26: 151 trials, 8×16, 74 sig ch)? Or different rule?

Meeting output: Locked claim language, first-round method list, implementation order, verification tasks.

Open Questions

Design questions:

1. Masked spans vs causal multi-step? (default: spans)
2. FHN parameter ranges—map interesting regime
3. Multi-variable observation for FHN: $(H, W, 2)$ or $(H, W, 1)$?
4. Prediction target space: raw patches or encoder tokens (stop-grad)?
5. Predictor depth: 2 vs 4 layers?

External verification:

6. **Array pitch** for 8×32 and 8×34—confirm from hardware docs or Zac
7. **Dev patient**: S26? Confirm
8. **Motor-focused crop**: exact boundaries from per-patient MFA timing? Patient-specific vs fixed?

Implementation order

1. Protocol scaffolding
2. Stage 1 synthetic stack
3. Model stack
4. Stage 2 training
5. Stage 3 evaluation
6. First-round execution
7. JEPA follow-up gate

Summary

Pretrain on **synthetic spatiotemporal grid dynamics**
(6 generators with controlled inductive biases)

↓

Adapt on **real response-locked trials** (weakly supervised)
with anti-leakage defenses

↓

Fine-tune a **lightweight head** (~2–5K params) per patient

↓

Answer: *Does structured synthetic dynamics improve motor-speech decoding from acute μECoG at <200 trials/patient?*

82 resolved decisions · 8 open questions · 7 first-round methods · 3 architecture families

Appendix: Priority Ablations

Ablation	Description
A. Pretext Objective	Masked spans (40–60%) vs causal multi-step (k=1,3,5) vs varied span lengths
B. Temporal Resolution	20 Hz (default) vs 40 Hz. Lock early
C. Patch Size	1×1 (default) vs 2×2 vs conv stem
D. Masking Ratio	30%, 50%, 70%. Interacts with temporal resolution
Tier 1 (gated on A > E)	
E. Generator Family	Single-family runs from the ladder
F. Freeze Level	Head+embed+LN vs head only vs full fine-tune
G. Architecture Family	Full expansion of conv-GRU / bridge / ViT
H. Spatial Pooling	Mean vs top-k vs learned attention
I. Source Weighting	Uniform vs similarity-weighted vs loss-balanced

Appendix: Parameter Budgets

Component	**Conv-GRU**	**Bridge**	**ViT**
Per-patient Conv2d	80	80	—
Per-patient patch embed	—	—	256
Spatial projection	—	576 + 256	—
Encoder	16,960	128	527,872
Predictor (BiGRU / Attn)	37,632	37,632	395,520
Decoder (discarded)	16,640	65	129
Coord PE	—	256	512
CE head	1,755	1,755	3,483
Total pretrain	~71K	~39K	~927K
Stage 3 trainable	~2.3K	~2.2K	~5.3K

All counts hand-computed—verify from prototype modules before running.

Appendix: Matching Statistics

Computed for each generator vs real uECOG (source patients only):

Statistic	**Method**
Temporal PSD	Welch (256-pt FFT, Hanning, 50% overlap), concatenated across trials. Evoked + residual
Spatial covariance	Exponential decay fit to Pearson r vs distance (mm). Evoked + residual
Amplitude distribution	Histogram of per-cell values; kurtosis + 95th percentile
Dead-mask prevalence	Fraction of zero/inactive cells
Event sparsity	Fraction of $\lvert z\rvert > 1.5$ cells per frame, per trial phase
Phase structure	Trial-averaged spatial variance time series; peak/baseline ratio
RT distribution	Mean, std, log-normal fit per patient

Coarse calibration: 1–2 parameters/generator to land inside broad acceptable bands (PSD below 20 Hz, spatial correlation 2–10 electrodes, kurtosis > 2, event sparsity 5–30%). Not optimization—prevents degenerate instantiation.

Appendix: References

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